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F78AB
Actuarial and Financial Mathematics

Chapter 1: Introduction

LEADERS IN **IDEAS** AND **SOLUTIONS**

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What is Financial Mathematics?

- It is a branch of mathematics dealing with **financial contracts** and **financial markets**.
- Can you think of any examples of financial contracts?
- What do we know about the way these contracts work?

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Cashflows

- In this course, we are particularly interested in the ***cash flows*** *(or payments)* associated with financial contracts
- We will think about the:
  - **Direction** of the cashflows (payments going out, or payment coming in?)
  - **Timing** of the cashflows
  - **Amount** of each cashflow
  - **Number** of cashflows being made

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Known Cashflows

- Some financial contracts involve cashflows where the amount or timing may not be known in advance. You can probably think of some examples of these.

- **In this course, we will consider only contracts where the cashflows are known in advance.**
- Later courses such as Life Insurance Maths and Risk Theory will build on the foundations of this course to introduce uncertainty into the payment amounts and timings – we shall ignore these for now (but note it is important to understand this course material well to make later courses accessible for you!)

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Reading - Chapter 1

- Chapter 1 includes
  - background information on financial contracts; and
  - Introduces the types of questions that actuaries and financial analysts might help to answer in relation to such contracts, not all of which are in the scope of this course

The **most useful** information in Chapter 1 is the introduction of the **timeline –** ***using and understanding timelines is the key to success in this course.***

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Introduction to timelines

1. You open a bank account and deposit £1,000. At the end of one year, the account is closed and the bank pays you £1,050.

2. You buy shares in British Airways which cost you £10,000. You hold onto the shares for 10 years; and receive a £400 dividend payment every six months during that period. After 10 years you sell the shares for £13,000.

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Next chapter...

Chapter 2: Interest and Present Value

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